

Unemployment Insurance, Wage Pass-Through, and Endogenous Take-Up

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*Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The research in this paper does not use any confidential Census Bureau information.

This Talk

How do (reservation) wages & UI take up respond to increase in UI benefits?

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Three main points

- 1 The wage and reservation wage response to Covid Acts was fairly modest... though the composition of UI collectors did change.
- 2 The take up rate of UI benefits went up
- 3 Using a simple model: make the case that these findings are related

Preview

- Using BAM* (2014-2022): random samples of UI paid and denied claims
 - Reservation wage is fairly insensitive to UI benefit: elasticity ≈ 0.014
 - Composition effect pulls res wage $\downarrow$, behavior pushes res wage $\uparrow$

★ Unemployment Insurance (UI) Benefit Accuracy Measurement (BAM)

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- Using BAM* (2014-2022): random samples of UI paid and denied claims
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 - Composition effect pulls res wage $\downarrow$, behavior pushes res wage $\uparrow$
- Using CPS (2014-2022): comparing post-COVID to pre-COVID
 - Share of UI eligible out of unemployed: 25% $\nearrow$ 80%
 - Share of UI collectors out of eligible: 30% $\nearrow$ 40%
 - Diff in wage after transition for UI eligible rel/to UI ineligible: $\uparrow$ by \$64 per week (9%)

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 - Diff in wage after transition for UI eligible rel/to UI ineligible: $\uparrow$ by \$64 per week (9%)
- Directed search model w/ OTJ search and endog. UI take up
 - Endog. collection dampens the response of wages to UI benefit increase

★ Unemployment Insurance (UI) Benefit Accuracy Measurement (BAM)

Literature

- **Impact of UI benefit and CARES Act on labor supply/search/spending:**
Cortes and Forsythe (JElneq2023, ILR Review2023); Forsythe, Kahn, Lange, and Wiczer (LE2022); Ganong, Noel and Vavra (JPubE2020); Ganong, Greig, Liebeskind, Noel, Sullivan and Vavra (2021); Finamor and Scott (EL2021); Boar and Mongey (2020); Petrosky-Nadeau and Valletta (2021); Michaud (2023); ...
- **Impact of UI benefit duration on employment:**
Hagedorn, Mitman and Manovskii (2016); Hagedorn, Karahan, Mitman and Manovskii (2019); Mitman and Rabinovich (JME2015, 2020); Chodorow-Reich, Coglianesi, and Karabarbounis (QJE2019); Acosta, Mueller, Nakamura and Steinsson (2023); ...
- **Endogenous take up rate:**
Auray, Fuller, and Lkhagvasuren (EEA2019); Auray, and Fuller (QE2020); ...

UI Eligibility and COVID Relief

Who is Eligible for Regular UI Benefit

Non-monetary criteria

Monetary criteria

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- Lost a job through no fault of their own
- Job quitters, new-entrants, and re-entrants are not eligible
- Haven't exhausted their benefits (typically 26 weeks)

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- Past earnings are sufficiently high (self-employed income doesn't count)
- **Benefits:** Date & state specific formula applied to employment income

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Claimants must also *maintain* eligibility

file (bi)weekly claims; be able/available to work; actively seek work; report any earnings; report any job offers; report to a local UI claims office; register for work w/ the State Employment Service; any other state requirements...

Changes to UI in response to COVID

Non-monetary criteria



Monetary criteria



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- Pandemic Emergency Unemp Comp (PEUC): Max # of weeks of UI: 26 ↗ 39

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Pandemic Unemployment Assistance (PUA) (backdated to Feb 2020)

→ Can apply after denial for regular UI (incl. PEUC)

- Unemployed for **any** self-certified COVID-related reason

Monetary criteria

- Self-employed ++ income qualifies
- Benefits: Minimum Disaster Unemployment Assistance (DUA)

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- Benefits: Minimum Disaster Unemployment Assistance (DUA)

Federal Pandemic Unemployment Compensation (FPUC)

- Add \$600/week (APR to JUL 2020)
- Add \$300/week (JAN to JUN/SEP 2021)

Monetary criteria

Evidence from BAM Data

UI Benefit Accuracy Measurement (BAM) Data

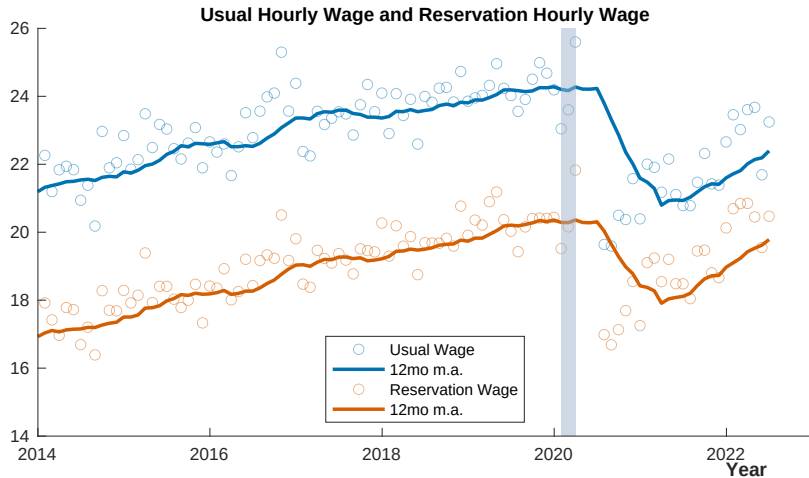
- Administrative data from the Department of Labor
- Used to assess accuracy of Regular UI paid/denied claims
- Representative samples of UI payments (paid claims) and disqualifying ineligibility determinations (denied claims)
 - Sampling based on fixed annual number of cases per state (roughly uniform monthly)
- Weekly data since 2002
 - Weekly unemployment benefits (paid claims)
 - Usual hourly wage
 - Reservation wage (lowest acceptable hourly wage)
 - Characteristics (age, gender, race, education, industry, etc.)
- Focus on 16-65 winsorized (1%) data, 2014-2022

Descriptive statistics (2014–2022)

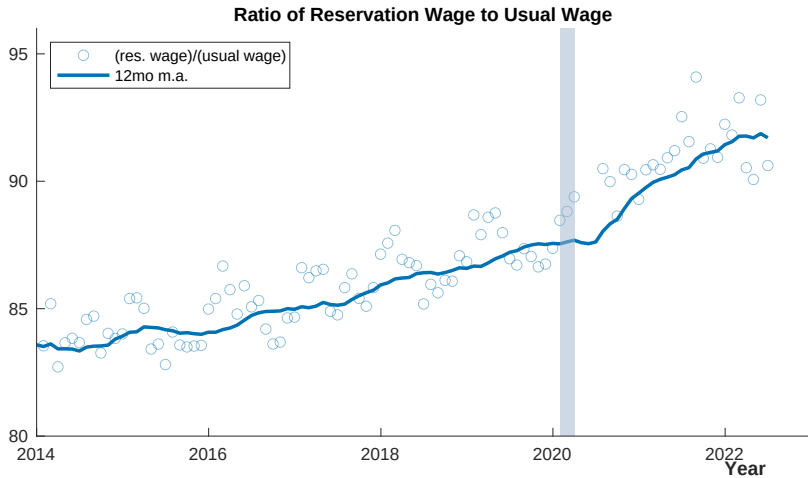
Sample	Paid Claims	Denied Claims	Monetary Denials
Age	41.24	38.23	36.81
Share Female	0.46	0.49	0.48
Share White	0.48	0.44	0.40
Share with College Degree	0.50	0.47	0.35
Share Manufacturing	0.23	0.19	0.15
Share Leisure & Hospitality	0.12	0.12	0.16
Usu. Hrly Wage (pre-Covid)	23.07	19.02	16.69
Usu. Hrly Wage (post-Covid)	20.89	18.57	16.57
Reservation Wage (pre-Covid)	18.91	16.11	14.39
Reservation Wage (post-Covid)	18.06	16.34	15.11
Weekly Benefit Amt (RUI)	354.75		
Observations	179,230	153,727	42,056

BAM: Paid Claims

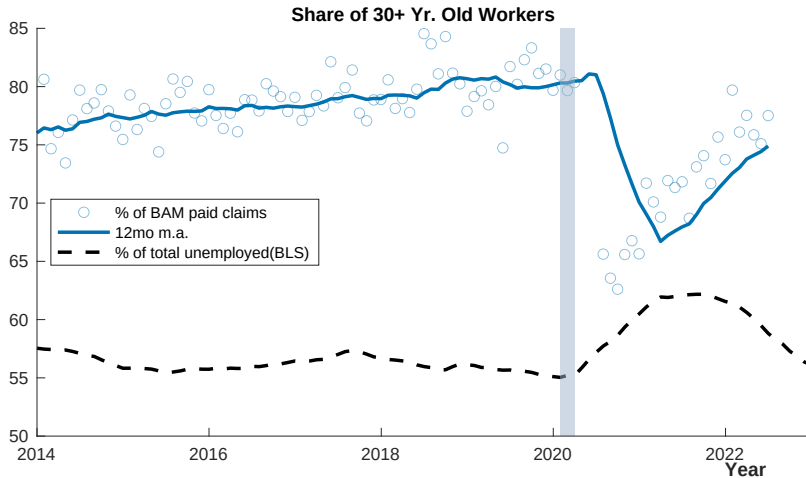
Usual Wage vs Res Wage – Paid Claims



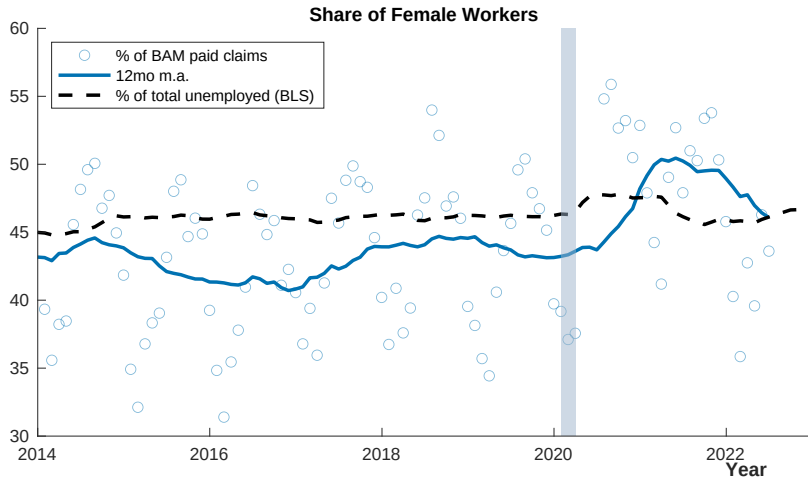
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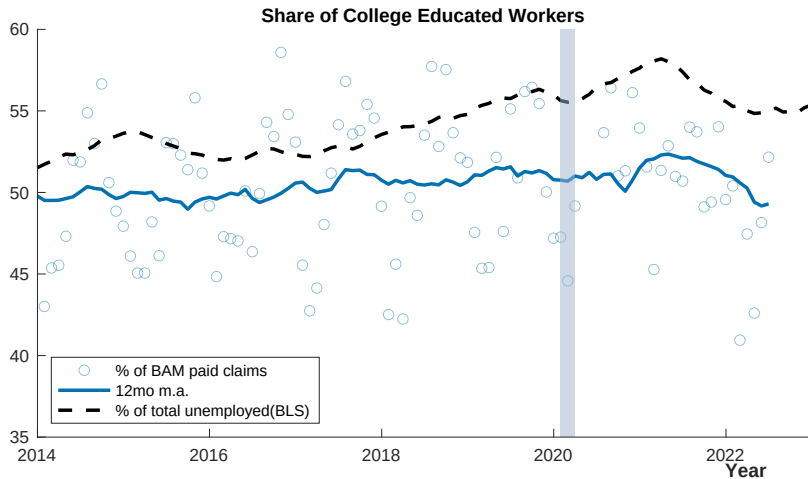
Composition – Paid Claims



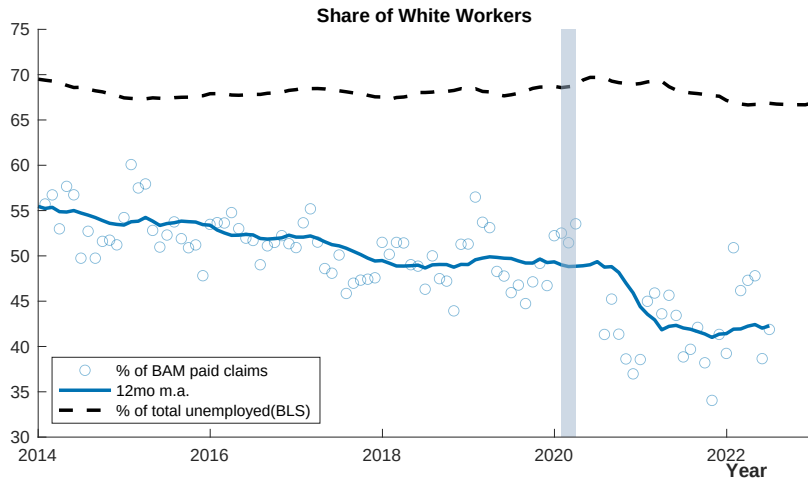
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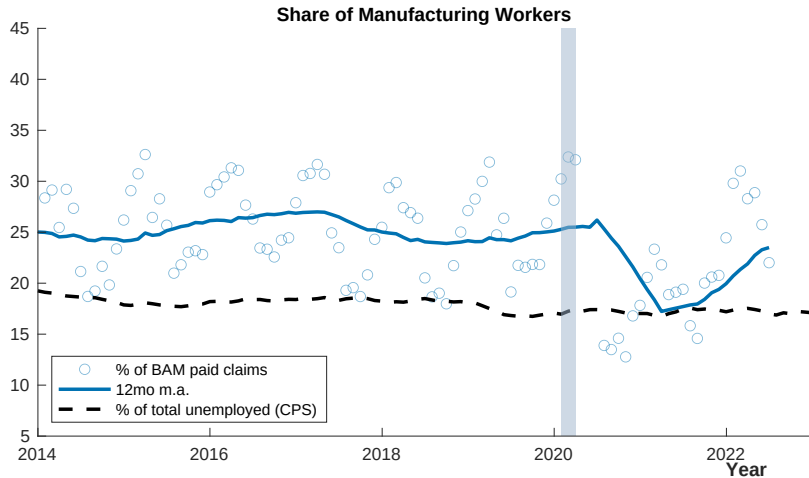
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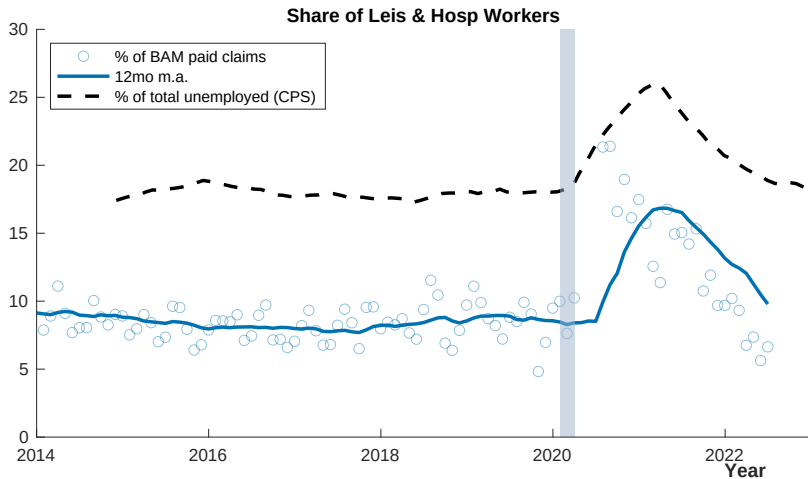
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Reservation Wage Elasticity – Paid Claims (2014–2022)

$$\log(w_{\text{res}}) = \beta_0 + \beta_1 \log(\text{Weekly UI Benefit}) + \beta_2 \log(w_{\text{usual}}) + \beta_3 X + \varepsilon$$

	$\ln(w_{\text{res}})$	$\ln(w_{\text{res}})$	$\ln(w_{\text{res}})$
$\ln(\text{Benefit})$	0.365*** (0.019)	0.025*** (0.007)	0.013* (0.007)
$\ln(w_{\text{usual}})$		0.800*** (0.020)	0.757*** (0.019)
PUA			0.011 (0.010)
U duration			-0.001** (0.000)
Age Cat./Gender/Edu/Race/Eth/NAICS/State dummies	No	No	Yes
Time dummies	No	No	Yes

Blinder-Oaxaca Decomposition:

- We want to understand the **change in reservation wage post 2019**
 - How much is due to change in behavior (possibly in response to COVID or policy)?
 - How much is due to change in composition?

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- We run the following regression Pre and Post COVID

$$\ln(w_{res}) = \beta X + \varepsilon$$

Vector of controls X :

Age groups; sex; education; race; 2-digit industry; log usual wage; state FE

Blinder-Oaxaca Decomposition:

- We want to understand the **change in reservation wage post 2019**
 - How much is due to change in behavior (possibly in response to COVID or policy)?
 - How much is due to change in composition?
- Decompose differences in mean $\ln(w_{res})$ pre and post COVID

$$\begin{aligned}\Delta E(\ln(w_{res})) &= E(X^{pre}) \Delta\beta && \leftarrow \text{due to "coefficients"} \\ &+ \Delta E(X) \beta^{pre} && \leftarrow \text{due to "endowments"} \\ &+ \Delta E(X) \Delta\beta && \leftarrow \text{due to "interactions"}\end{aligned}$$

Blinder-Oaxaca Decomposition: Paid Claims Pre-Covid vs 2020

	Decomposition	95% CI
Pre Covid	2.798***	[2.761,2.835]
2020 Expanded Benefit	2.714***	[2.656,2.771]
Difference	-0.085***	[-0.133,-0.036]
Explained Component (endowments)	-0.127***	[-0.166,-0.088]
Unexplained Component (coefficients)	0.040***	[0.014,0.066]
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- 2020 res wage **8.5 percent lower** than pre-Covid
- Composition change **pulls it down by about 13 percent**
- Coefficient change **pushes it up by 4.0 percent**

Blinder-Oaxaca Decomposition: Paid Claims Pre-Covid vs 2021

	Decomposition	95% CI
Pre Covid	2.798***	[2.761,2.835]
2021 Expanded Benefit	2.851***	[2.805,2.896]
Difference	0.052***	[0.027,0.078]
Explained Component (endowments)	-0.022**	[-0.040,-0.004]
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- 2021 res wage **5.2 percent higher** than pre-Covid
- Composition change **pulls it down by about 2.2 percent**

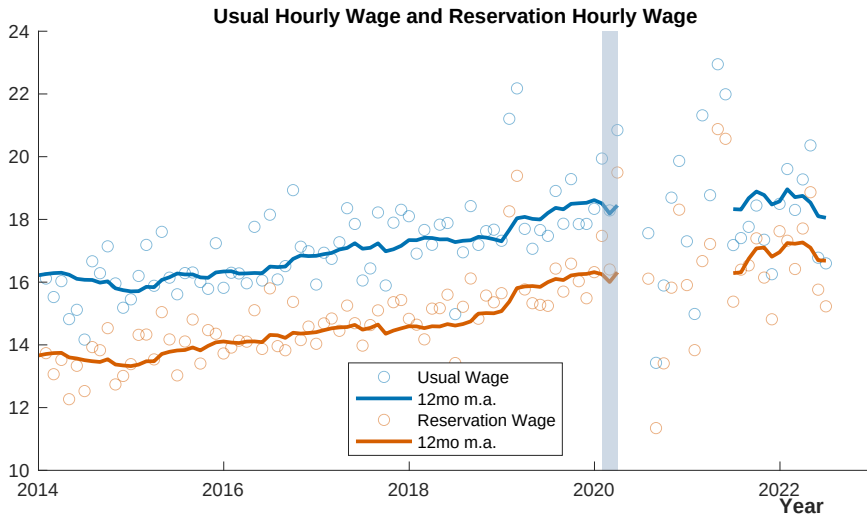
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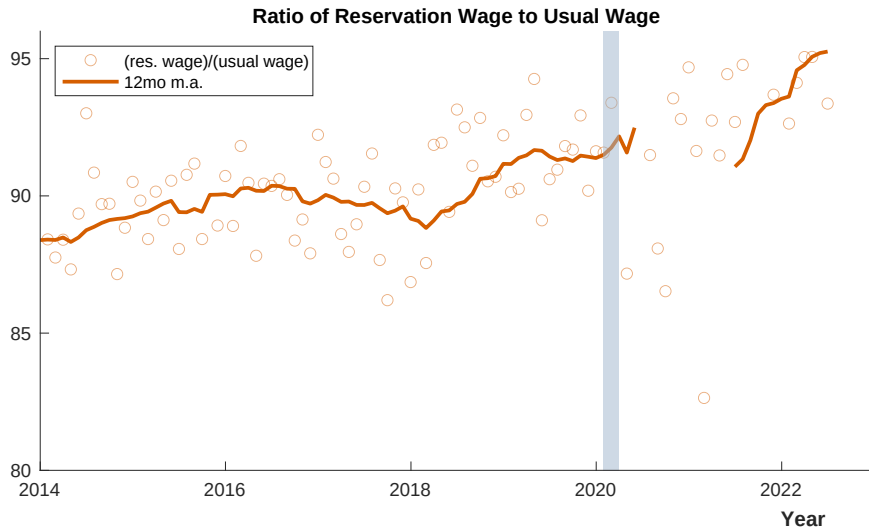
- 2021 res wage **5.2 percent higher** than pre-Covid
- Composition change **pulls it down by about 2.2 percent**
- Coefficient change **pushes it up by 7.3 percent**

BAM: Denied Claims

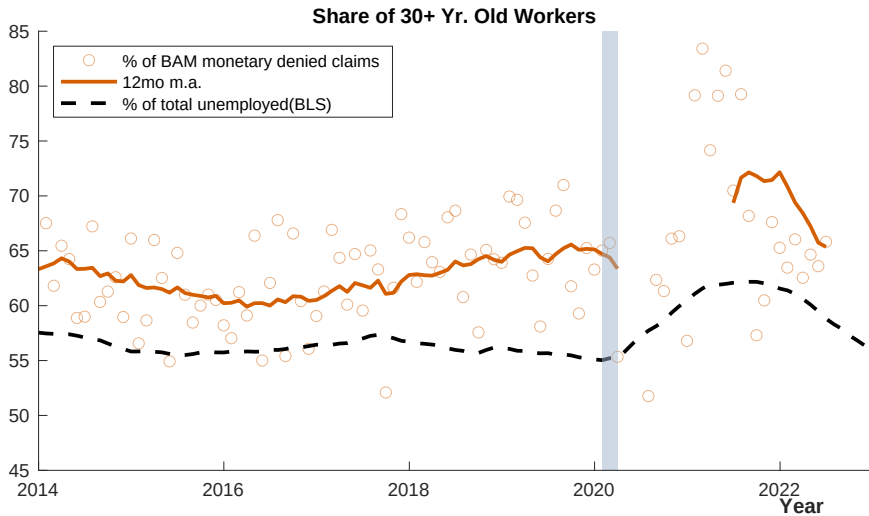
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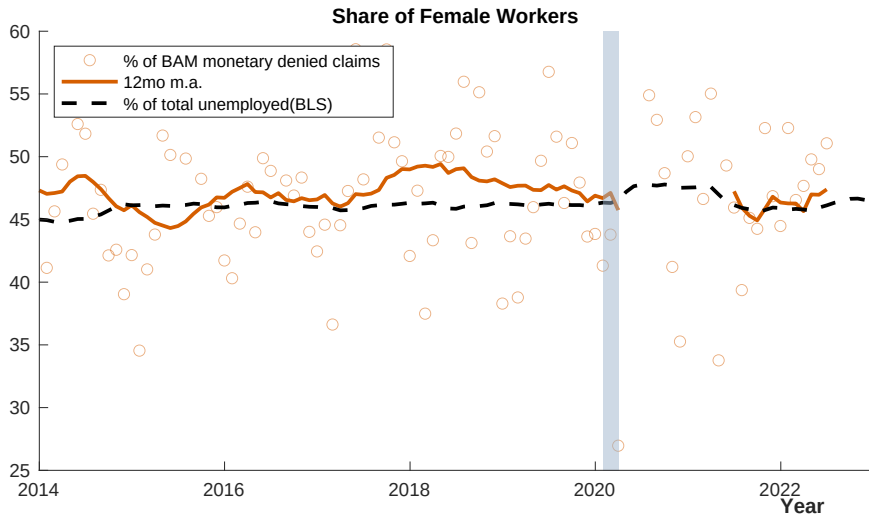
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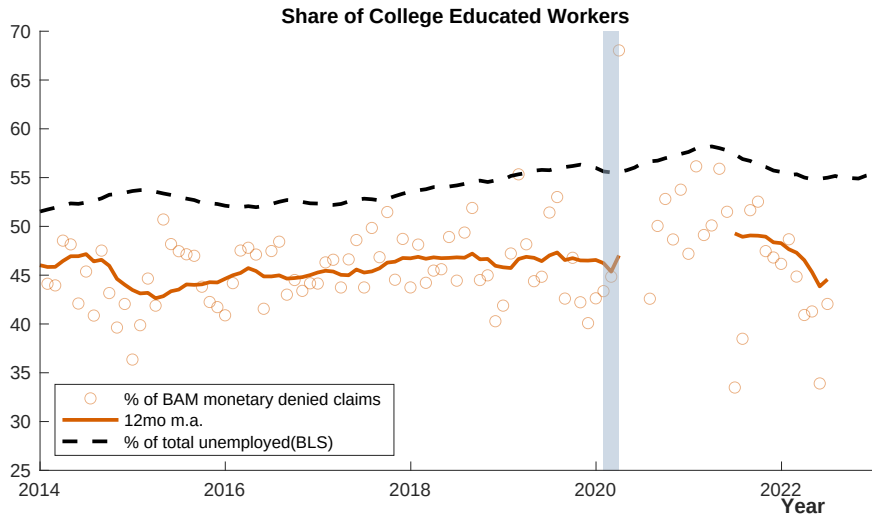
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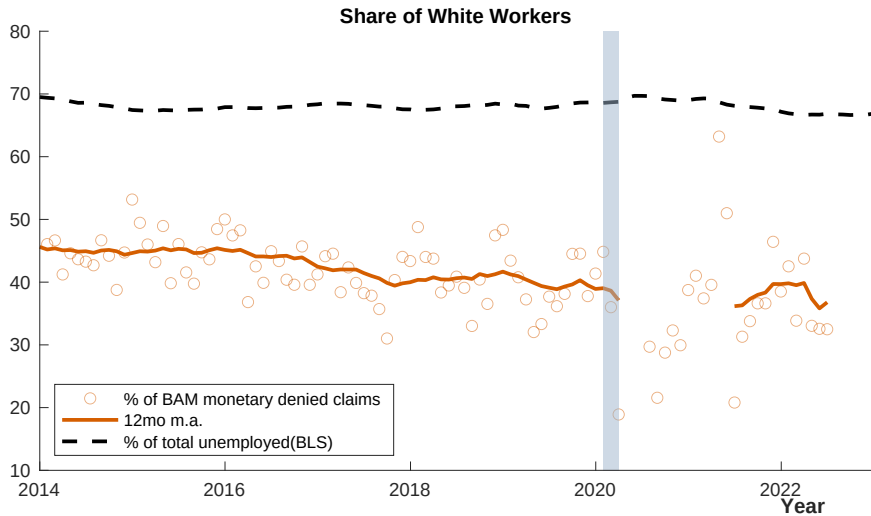
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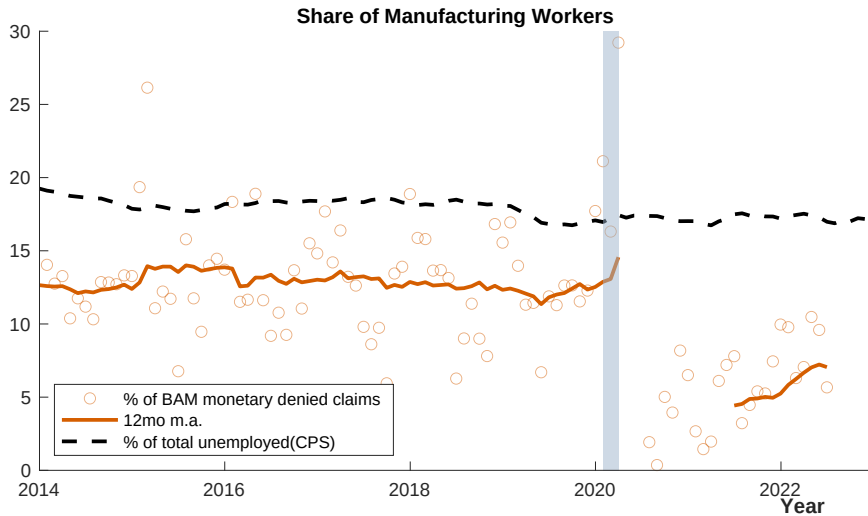
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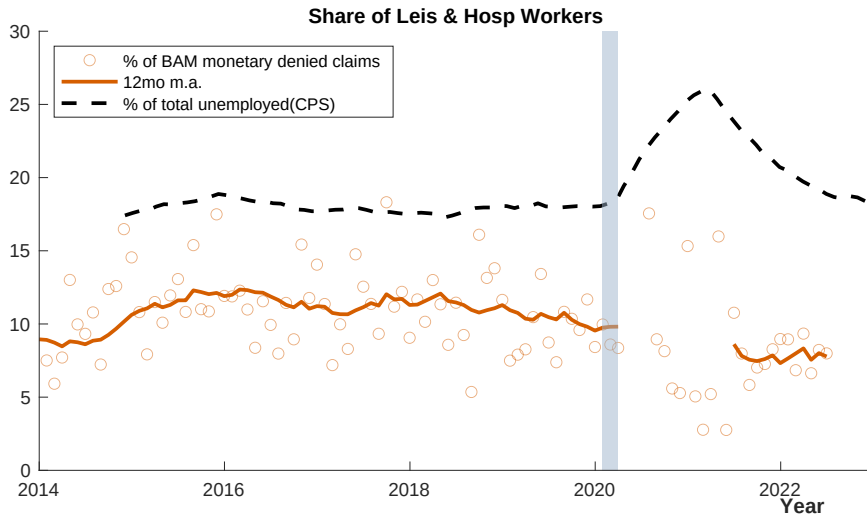
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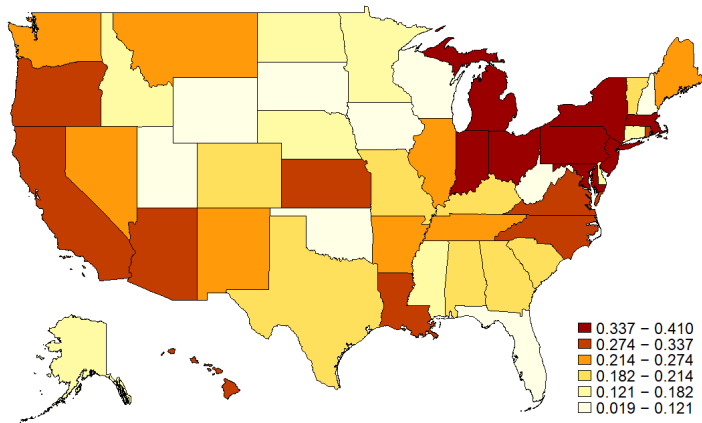


Empirical Design – Denied Claims

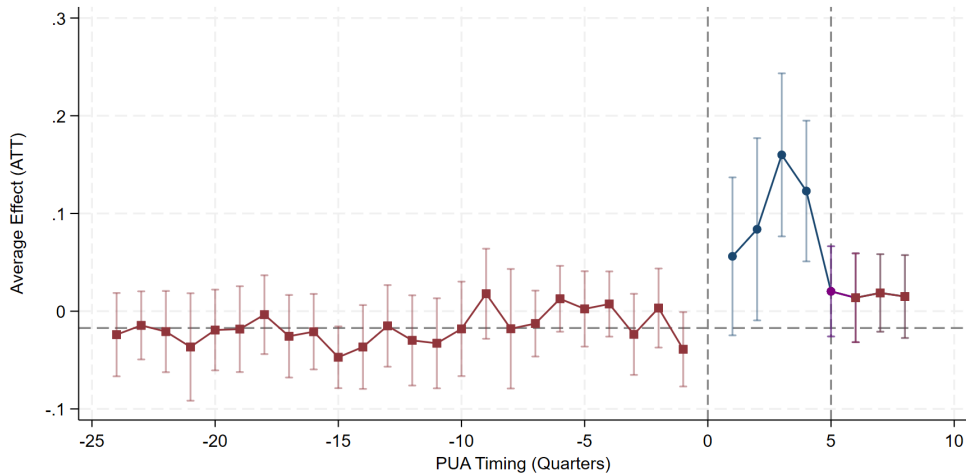
- Pandemic Unemployment Assistance (PUA) required ineligibility from Regular UI
- Incentive for low earnings people to apply and be denied for R-UI
 - ⇒ Many denied claimants in 2020 and 2021 expected to qualify for PUA (and FPUC)
- Exploit variation in state compliance with PUA (next slide)
 - ⇒ High State: share of PUA claims in total claims above median
- Use variation in state compliance to estimate how expected benefits in ‘high states’ increased reservation wage:

$$\log(w_{\text{res}}) = \beta_0 + \beta_1 \mathbb{I}(\text{PUA}) + \beta_2 \mathbb{I}(\text{High States}) + \alpha \mathbb{I}(\text{PUA}) \times \mathbb{I}(\text{High States}) + \beta_3 X + \varepsilon$$

Share of PUA claims in total claims



Diff in Diff Results α



Continuous Treatment Specification

- Use continuous treatment intensity to get a measure of elasticity of res wage to UI benefit
- PUA_{it} is qtlly average of state's share of weekly PUA/unemployed pop.
- Control for state-level qtlly average of weekly COVID deaths/100k and state-level unemployment rate.
- Look at difference in α_t from PUA-only period vs PUA + FPUC.

$$Y_{it} = \lambda_t + \alpha_t PUA_{it} * period_t + \delta Covid_{it} + \gamma Urate_{it} + \beta X_{it} + \varepsilon_{it}$$

Interact with quarterly time dummies:

$$Y_{it} = \lambda_t + \alpha_t PUA_{it} * period_t + \delta_t Covid_{it} * qtr_t + \gamma_t Urate_{it} * qtr_t + \beta X_{it} + \varepsilon_{it}$$

Continuous Treatment Specification

	$\ln(\tilde{w})$	$\ln(\tilde{w})$	$\ln(\tilde{w})$
PUA	0.0231 (0.59)	0.0289 (0.84)	-0.00759 (-0.20)
PUA + FPUC	0.0992** (3.34)	0.108*** (3.79)	0.116* (2.67)
Difference	0.076	0.079	0.124
Qtly Controls	No	Yes	No
Qtly Controls x Time	No	No	Yes
N	33868	33868	33868

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

► Quarterly Time-series

Elasticity of Res Wage to UI Benefit

- Moving from PUA claim share of 0 to 1 is associated with an α_t increase in reservation wage.
- Difference in PUA vs PUA + FPUC coeff. is change in coeff. from \$300 FPUC supplement.
- Assume average DUA payment of \$176 for PUA claimants then benefits increase 170% to \$476.
- Implies elasticity of res wage to UI benefit of $7.6/170 = .045$ to $11.6/170 = .068$

Take away from BAM data

- Elasticity of res wage w/r to benefit ≈ 0.014
- Composition of claimants pulled res wage down by ≈ 2 to 10%
- Extra benefits pushed res wage up by ≈ 4 to 7%
- Res wage of denied claims increased in high PUA states when benefits were elevated
- Implies an elasticity of res wage w.r.t. UI benefit of .045 to .068

Evidence from CPS Data

Data: Survey Structure and Data Requirements in CPS

D	J	F	March	A	M	J	J	A	S	O	N	D
			1	2	3	4						
		1	2	3	4							
1	1	2	3	4								5
	2	3	4									
			5	6	7	8						
		5	6	7	8							
5	5	6	7	8								
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- 4 months in – 8 months out – 4 months in
- Weekly earnings asked in **outgoing rotations**
- Can assess non-mon eligibility in each month
- Mon eligibility? Need prev year income

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		3	3	4								
		4	4									5
5	5 6	5	5	6	7	8						
		6	6	7	8							
		7	7	8								
		8	8									

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- Restrict sample to only those who have interview(s) in March

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“Did receive UI benefit last year?”

- Use this to find UI collectors, retroactively

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First wage after Unemp spell?

Need a $U \rightarrow E$ before
outgoing rotations

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2020 COVID relief starts in April

Very limited sample in 2020

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2021 COVID relief has more coverage

Better sample in 2021

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Measuring UI Benefits Eligibility

- Regular Unemployment Benefits

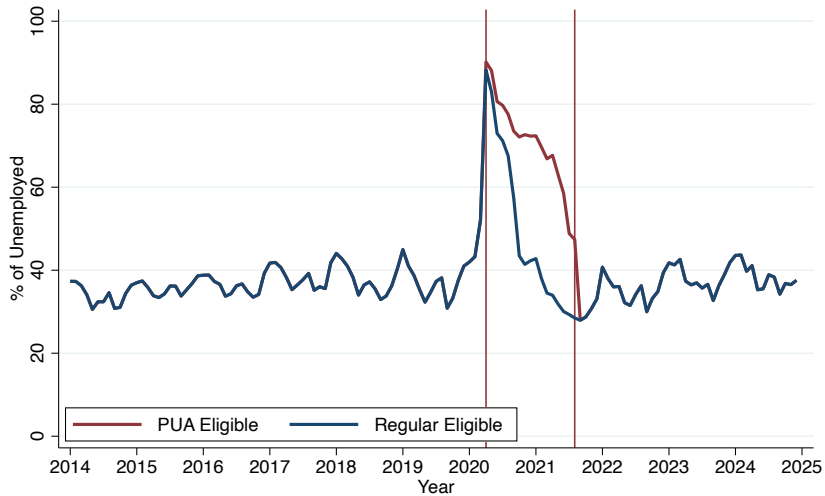
- Experienced unemployed who lost job
- Unemployed < 26 weeks
- Use Ganong et al. (2020) 'ui-calculator' to assess monetary eligibility

- COVID Relief Period

- 'Unable to work/look-for-work due to COVID' (CPS questions)
- We ignore UI duration
- Self-employment income qualifies (goes to 'ui-calculator')
- Alt: evidence that work was interrupted in any way by COVID-19
basically, evidence of any positive income (incl self employment)

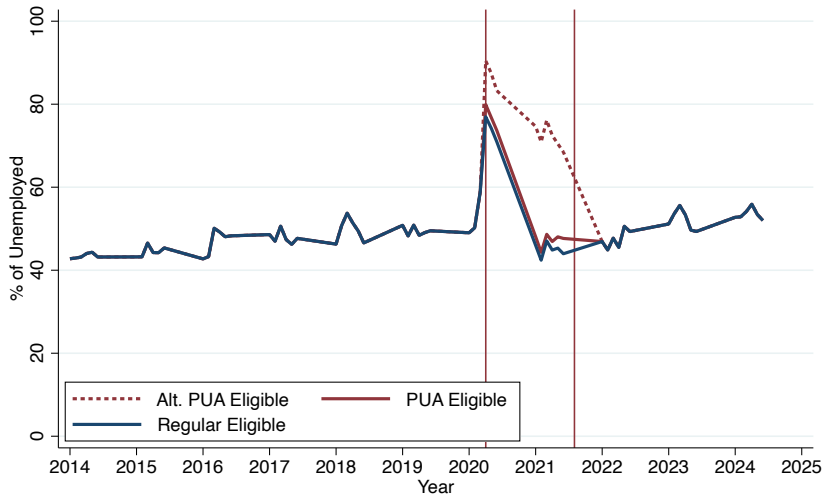
Share of Unemployed Who are Eligible for UI

► Composition Changes



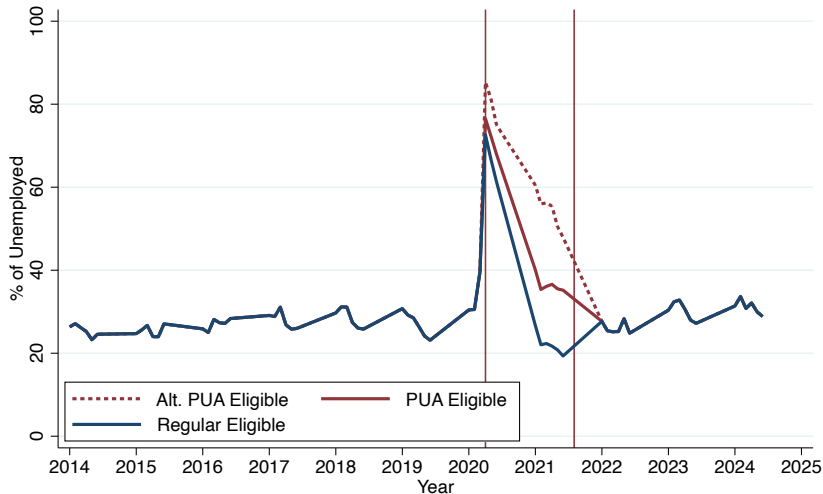
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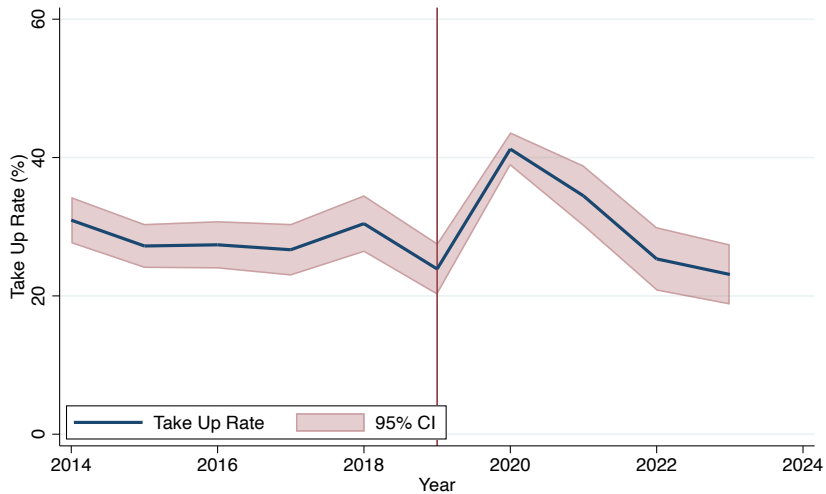
UI Take Up Rate

- Use March CPS question on whether UI income received in prev year
- Probit regression to estimate the propensity to claim pre-2020
 - on sample of eligible unemployed
 - as function of characteristics (age, sex, education, race, occupation, and industry)
 - duration of spell, past earnings
 - Once excluding benefit amounts and once including benefit amounts
- Use coefficients to predict take up rate in 2020 and 2021

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- Use coefficients to predict take up rate in 2020 and 2021
- Goal:
 - How much of the change in take up rate is due to change in composition?
 - How much of the change in take up rate is due to change in policy?

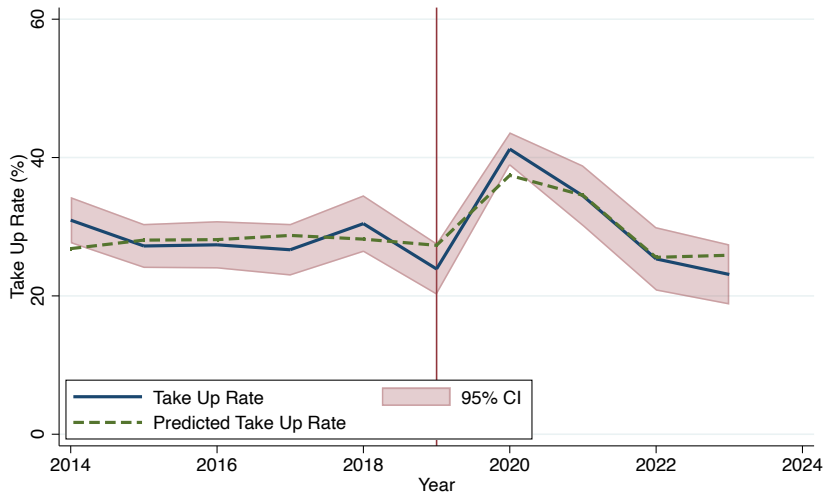
UI Take Up Rate



UI Take Up Rate



UI Take Up Rate



Earnings Following Transition to Employment

- Reported weekly wages at “outgoing rotations” (interview 4 and 8)
- Observe transition to employment in 2nd, 3rd, or 4th (6th, 7th, or 8th) interviews

Earnings Following Transition to Employment

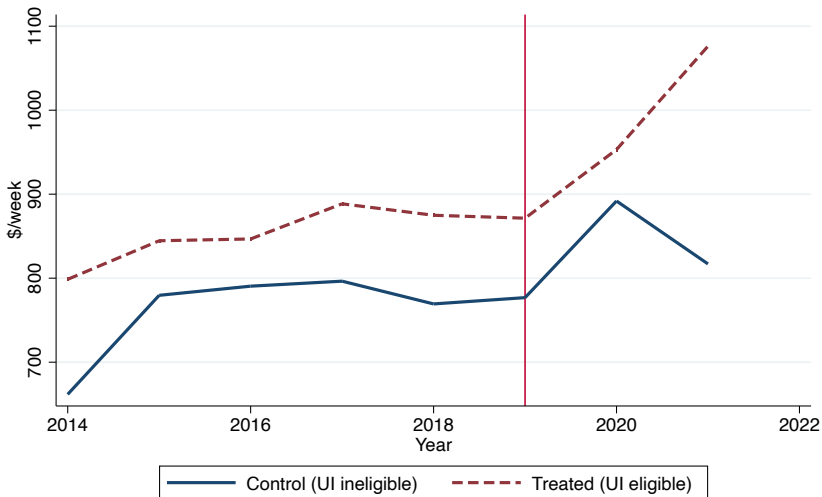
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- Did it go up after 2020?

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- Did it go up after 2020?
- To answer, run the following Diff in Diff regression

$$w = \alpha_0 \mathbb{I}(\text{eligible}) + \alpha_1 \mathbb{I}(\text{COVID}) + \gamma \mathbb{I}(\text{eligible}) \mathbb{I}(\text{COVID}) + \beta X + \varepsilon,$$

1st Wage after Transition: UI Eligible vs UI non-Eligible



DiD Results: Eligible Wage Premium pre vs COVID

	Include 2020		Exclude 2020	
	(1)	(2)	(3)	(4)
Δ wage premium (\$)	91.00*** (33.81)	-4.437 (41.08)	73.34** (30.08)	64.34** (26.84)
year fixed effect	No	Yes	No	Yes
<i>N</i>	7,523	7,523	6,719	6,719

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Data from the CPS. Additional controls in all regressions are:
gender, age, education, race, industry and occupation.

Take away from CPS

- UI take up rate went up because of higher benefits
- Wage premium of eligible vs non-eligible is positive, and went up
... well... if we ignore 2020!

Taking Stock

- Reservation wage responds mildly to UI benefit
- Gap in 1st wage after transition to employment bwn eligible & ineligible positive and increased in 2021
- UI take up rate increased significantly – explained by increase in benefit levels

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- Next: a directed search model w/
 - OTJ search: gives us 1st wage after transition
 - endogenous UI take up decision
 - Notion of eligibility and UI duration (not today)

A Directed Search Model with Endogenous UI Collection

Economic Environment: Directed Search

- Preferences: $\sum_{t=0}^{\infty} \beta^t v(c_t)$, $0 < \beta < 1$
- Endowments: 1 unit of time per period
- Technology: produce $y (= 1)$ when matched
- OTJ Search: employed workers search with prob. $\lambda_e < 1$
- Unemployed workers search with prob. 1
- UI collectors and non-collectors (next slide)
- Free-Entry of firms

UI Takeup

- Inspired by Auray, Fuller and Lkhagvasuren (2019)
- Utility cost of collecting UI benefit: $\varepsilon \in [\underline{\varepsilon}, \bar{\varepsilon}]$ from $F(\varepsilon)$
- Draws are i.i.d. across workers and across unemployment spells
- Each draw stays with the worker until he exits unemployment
- Collectors receive b and incur utility cost ε
- Non-collectors receive $d < b$, with $v(b) - \bar{\varepsilon} < v(d) < v(b) - \underline{\varepsilon} < v(b)$

Economic Environment: Timing

Each period consists of 4 stages:

- 1 Exogenous separation with probability δ
- 2 Search: employed and unemployed workers search
- 3 Matching: workers find job with probability $p(\theta)$ and (θ = market tightness)
 - Vacancies fill with prob. $q(\theta) = p(\theta)/\theta$
- 4 Production and consumption:
 - Employed workers produce output and consume w
 - Unemployed consume b or d

Individuals' Search Value

- Search value for an individual with current utility $x = V$:

$$R(V) = \max_{x \in X} p(\theta(x))(x - V)$$

$m(V)$: optimal submarket

$\tilde{p}(V) = p(\theta(m(V)))$: Job Finding Prob in optimal submarket

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- Note that

$$R'(V) = -\tilde{p}(V) < 0$$

- Search value is decreasing in current promised utility

Value of Unemployment and decision to collect UI

- For UI non-collectors:

$$U^N = v(d) + \beta \left\{ U^N + R \left(U^N \right) \right\}$$

- For UI collectors:

$$U^C(\varepsilon) = v(b) - \varepsilon + \beta \left\{ U^C(\varepsilon) + R \left(U^C(\varepsilon) \right) \right\}$$

- Collection decision:

$$U(\varepsilon) \equiv \max \left\{ U^N, U^C(\varepsilon) \right\}$$

- value prior to realization of ε

$$U \equiv \int_{\underline{\varepsilon}}^{\bar{\varepsilon}} U(\varepsilon) dF(\varepsilon)$$

Firm's Problem

- Firm who matches with a worker in sumarket $x = V$, offers contract (w, V')
 - wage: w
 - the continuation value (promised utility): V'
- The value function of the firm:

$$J(V) = \max_{\{w, V'\}} \left\{ y - w + \beta \left[(1 - \delta)(1 - \lambda_e \tilde{p}(V')) J(V') \right] \right\}$$

$$V = v(w) + \beta \left\{ \delta U' + (1 - \delta) [V' + \lambda_e R(V')] \right\}$$

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- Free entry implies

$$\kappa \geq q(\theta(x)) J(x)$$

Few Technical Results: market tightness

- Firm value function $J(x)$ is decreasing and concave
- Free entry condition

$$\kappa = q(\theta(x))J(x)$$

implies that

Market tightness $\theta(x)$ is decreasing in promised value x

Few Technical Results: optimal search

- Recall: optimal search problem an individual with current utility $x = V$:

$$R(V) = \max_{x \in X} p(\theta(x))(x - V)$$

$m(V)$: optimal submarket

$\tilde{p}(V) = p(\theta(m(V)))$: Job Finding Prob in optimal submarket

- Workers always search for higher promised value: $m'(V) > 0$ and $m(V) \geq V$
- It follows that job finding rate is decreasing in current promised value V

$$\tilde{p}'(V) < 0$$

Few Technical Results: value of being unemployed

- For collectors

$$U^C(\varepsilon) = v(b) - \varepsilon + \beta \left\{ U^C(\varepsilon) + R \left(U^C(\varepsilon) \right) \right\}$$

$U(\varepsilon)$ is decreasing in ε .

- For non-collectors

$$U^N = v(d) + \beta \left\{ U^N + R \left(U^N \right) \right\}$$

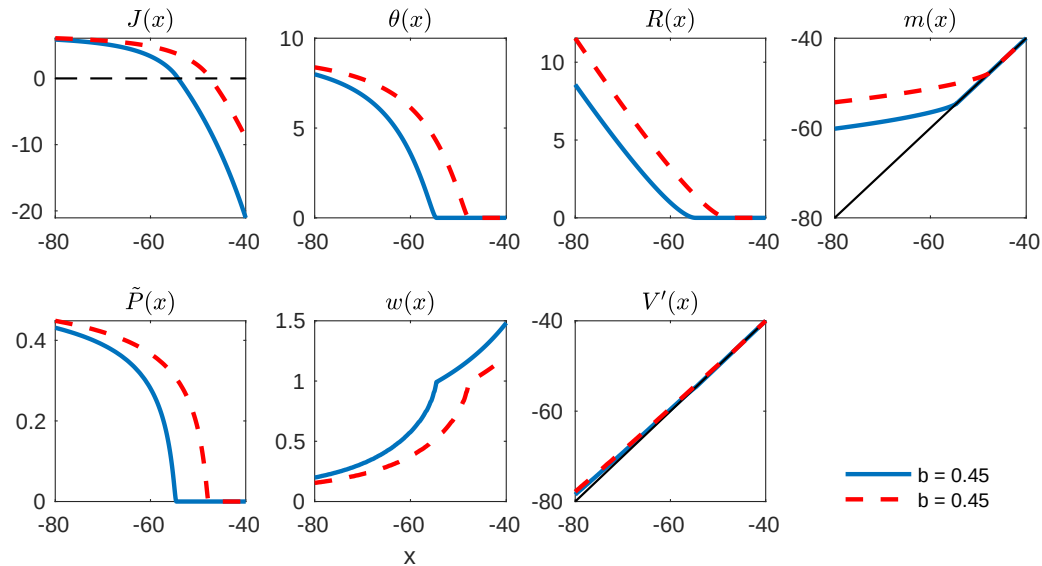
- Therefore, there is a cutoff $\varepsilon^* = u(b) - u(d)$ such that $U^C(\varepsilon^*) = U^N$
- Everyone with $\varepsilon < \varepsilon^*$ will collect
- Non-collectors have the lowest value

Few Technical Results: job finding rates and wages

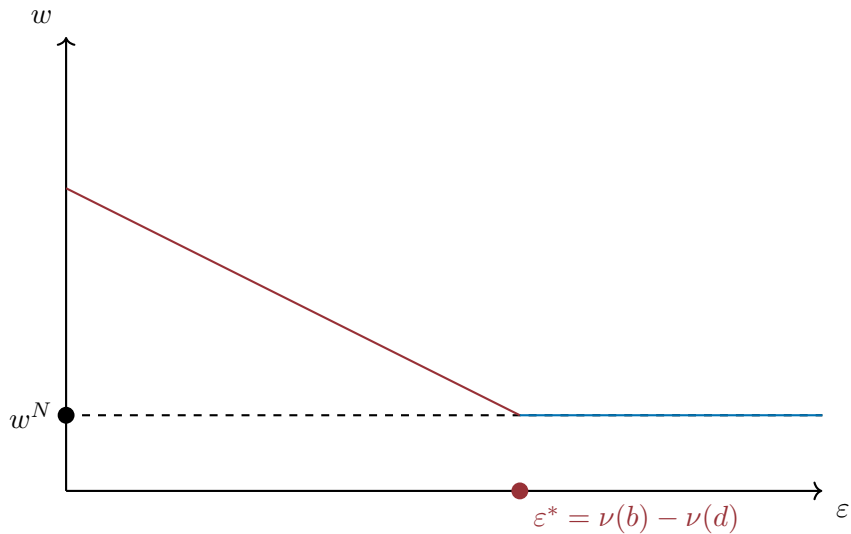
- Recall that
 - Value of being unemployed is decreasing in ε
 - Optimal job finding rate $\tilde{p}(V)$ is decreasing in V
- Therefore, individuals with high ε search in markets with high job finding rate
- There is a trade off: they go for lower wages
- Non-collectors search in sub-market with lowest wages.

Comparative Statics

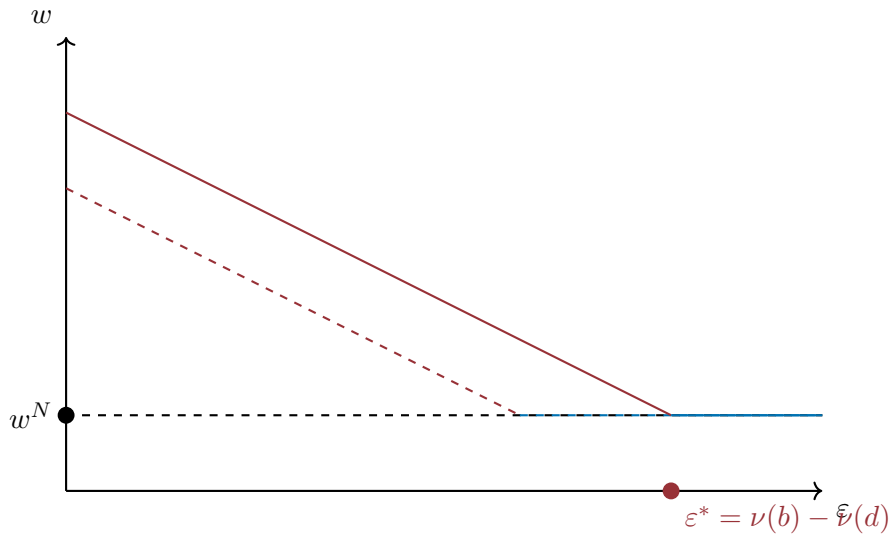
- Increasing UI benefit b has two effects:
 - It increases value of being unemployed today: $U(\varepsilon)$
 - It increases the expected value of being unemployed in the future $U \equiv \int_{\underline{\varepsilon}}^{\bar{\varepsilon}} U(\varepsilon) dF(\varepsilon)$
- The former make being unemployed more valuable
 - $\Rightarrow$ makes workers more selective $\Rightarrow$ search for higher values/wages
- The latter make being employed more valuable
 - $\Rightarrow$ makes workers less selective $\Rightarrow$ search for lower values/wages
- First effect always dominates (because of discounting)



The effect of higher UI benefit



The effect of higher UI benefit



Parameterization

- Preference:

$$\nu(c) = (c^{(1-\sigma)} - 1)/(1 - \sigma), \quad \sigma = 2, \quad \beta = 0.996$$

- Matching technology:

$$p(\theta) = \theta(1 + \theta^\gamma)^{-1/\gamma}, \quad \gamma = 0.5$$

- OTJ search: $\lambda_e = 0.25$

- Exogenous separation rate: $\delta = 0.0145$

- Vacancy cost: $\kappa = 0.32$

- $\varepsilon \sim U[0, 9.25] \leftarrow 31.0\%$ take up rates

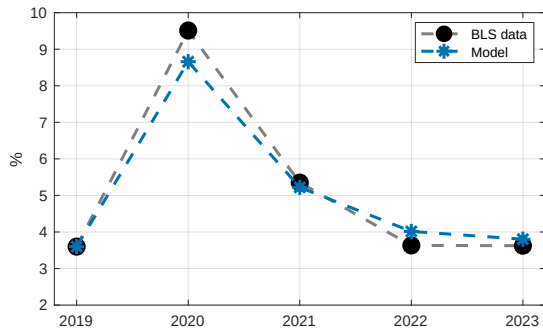
- $d = 0.2, b = 0.45$

Simulating a Covid-type Recession

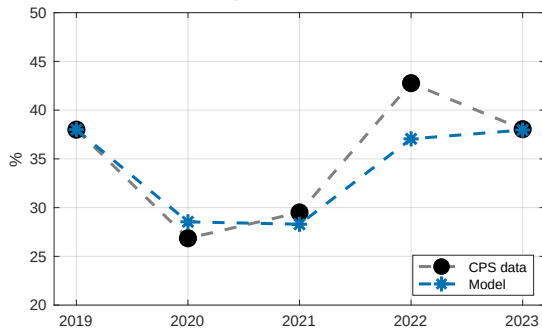
- Break up random matches to raise unemployment rate up to 15%
- Make all unemployed eligible for UI and increase UI compensation ($b - d$ doubles for 6 months, back to zero for 4 months, $b - d$ up by 1.5 for 9 months)
- Create recession by increasing vacancy cost by 50% (can also change match efficiency with similar results):
 - Match the path of JFP/U-rate over time
- Interested in wages (claimants rel/to non-collectors and eligible vs ineligible) and UI take up rate
- Experiment: exogenous take up decision (fix the cutoff cost for collectors)

Simulation Results

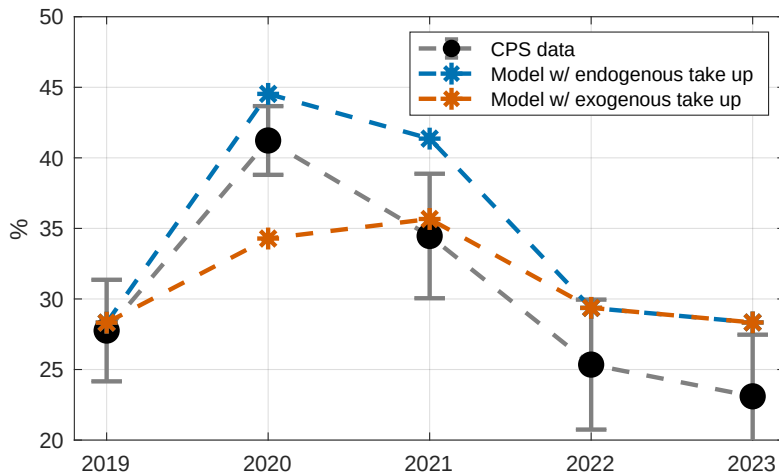
Unemployment Rate, Model vs. Data



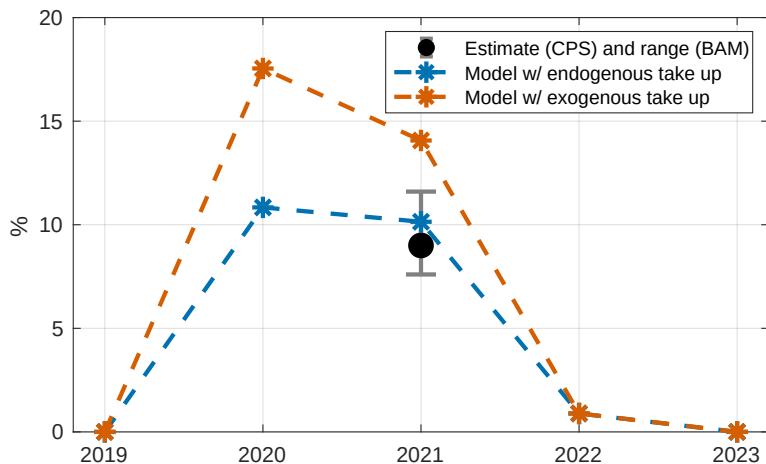
Job-Finding Rate, Model vs. Data



Collection Rate (% of eligible): Endog. vs. Exog. Take Up



Collector Wage Premium: Endog. vs. Exog. Take Up



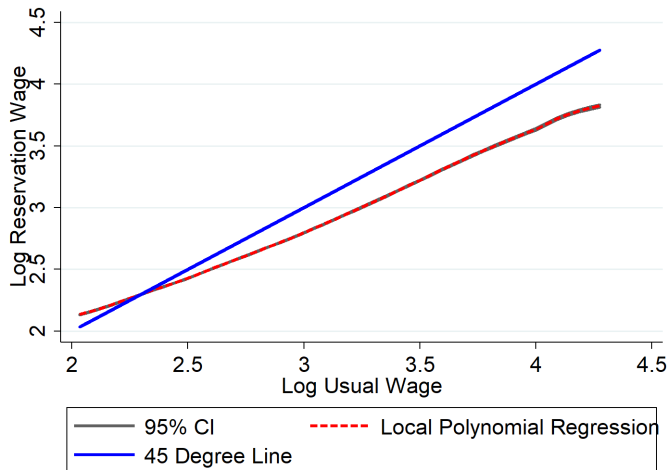
Concluding Remarks

- Using BAM, we showed:
 - Composition of UI collectors changed during COVID
- Using CPS, we showed:
 - Significant benefit-driven increase in UI take up during COVID Acts
- Using CPS & BAM, we showed:
 - Modest (reservation) wage response to rise in UI benefit
- Using a simple model of directed search OTJ:
 - Endog. take up rate dampens wage response to benefits

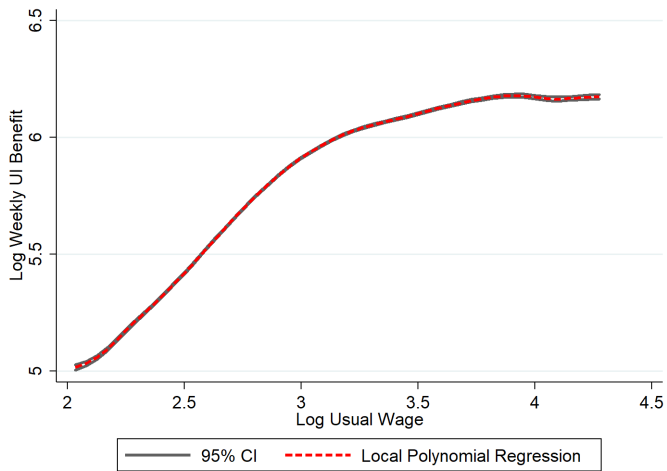
Reasons for not applying for UI benefits (2018 UI Supplement)

Mains reasons	Percent	Cum.
do not need the money or do not want the hassle	19.42	19.42
negative attitude about unemployment Insurance	2.45	21.87
do not know about unemployment Insurance or or do not know how to file	5.94	27.81
problems with application process	2.05	29.86
plan to file soon	4.68	34.55
constraints on accepting employment	25.07	59.62
expect to start working soon	19.56	79.18
all other reasons / reason not provided	20.82	100.00

Usual Hourly Wage and Reservation Wage – Paid Claims

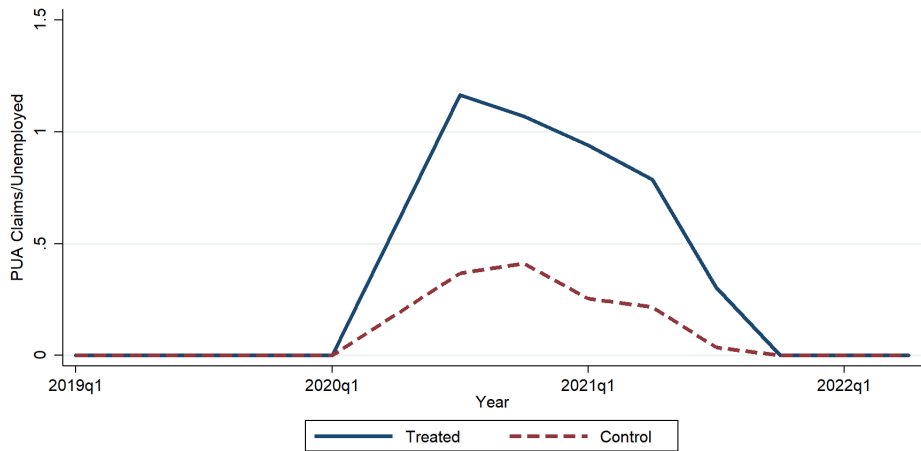


Weekly Benefit and Usual Hourly Wage – Paid Claims



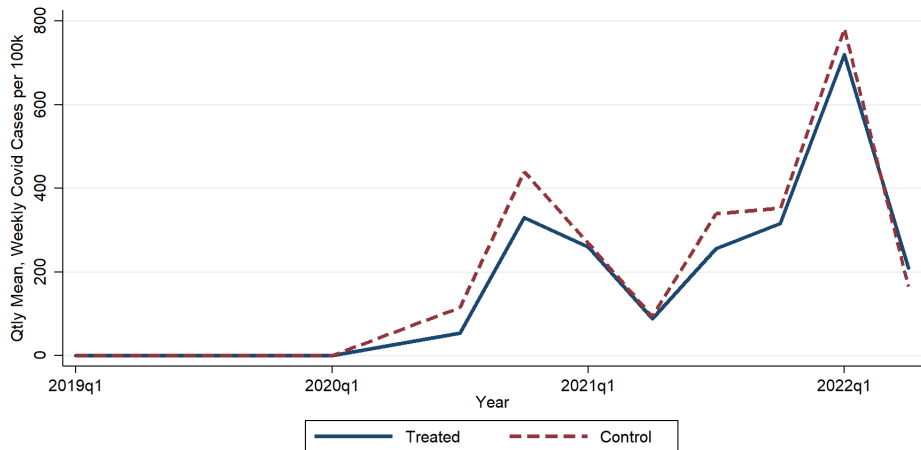
Quarterly Time Series of “Treatment” vs “Control” States

PUA Claims as Share of Total Unemployment



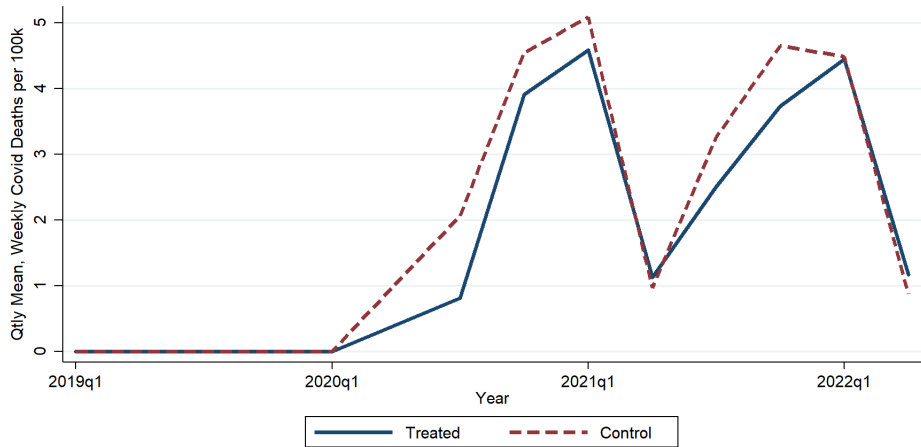
Quarterly Time Series of “Treatment” vs “Control” States

Covid Cases/100k Individuals



Quarterly Time Series of “Treatment” vs “Control” States

Covid Deaths/100k Individuals



Quarterly Time Series of “Treatment” vs “Control” States

Unemployment Rate

